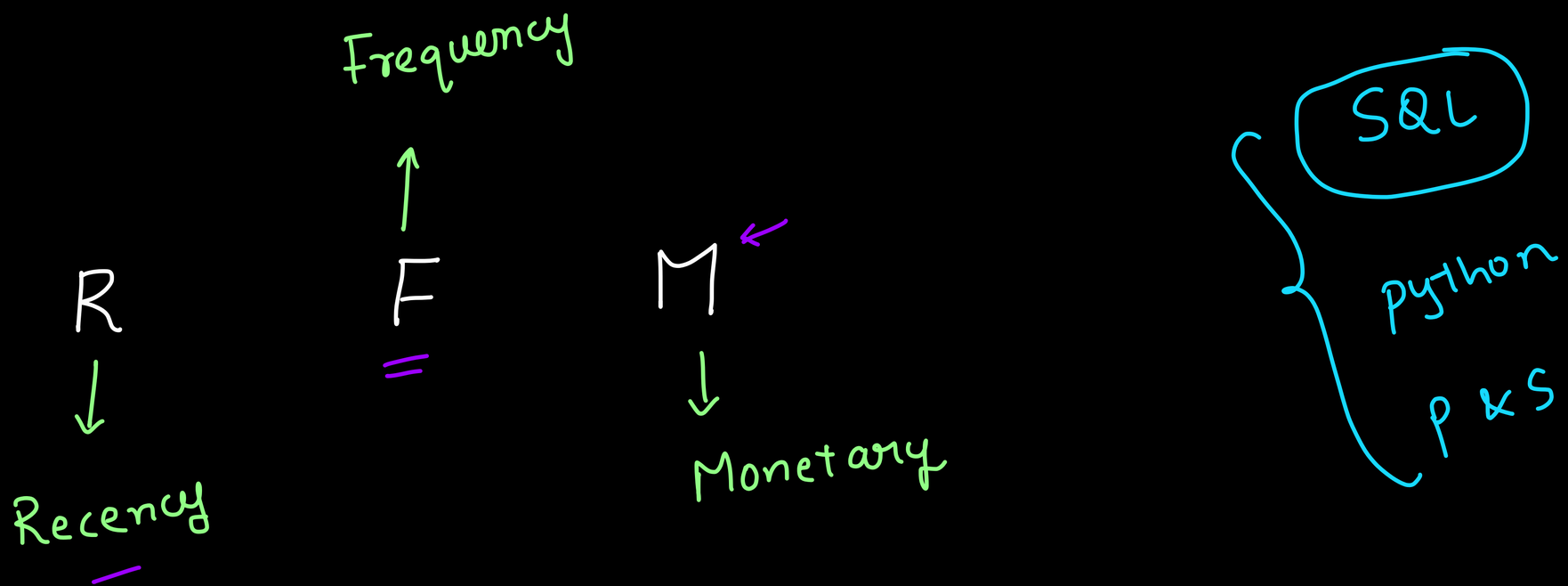
Agenda

* Customer Segmentation

  - RFM Analysis for a chain of retail stores

Assignments



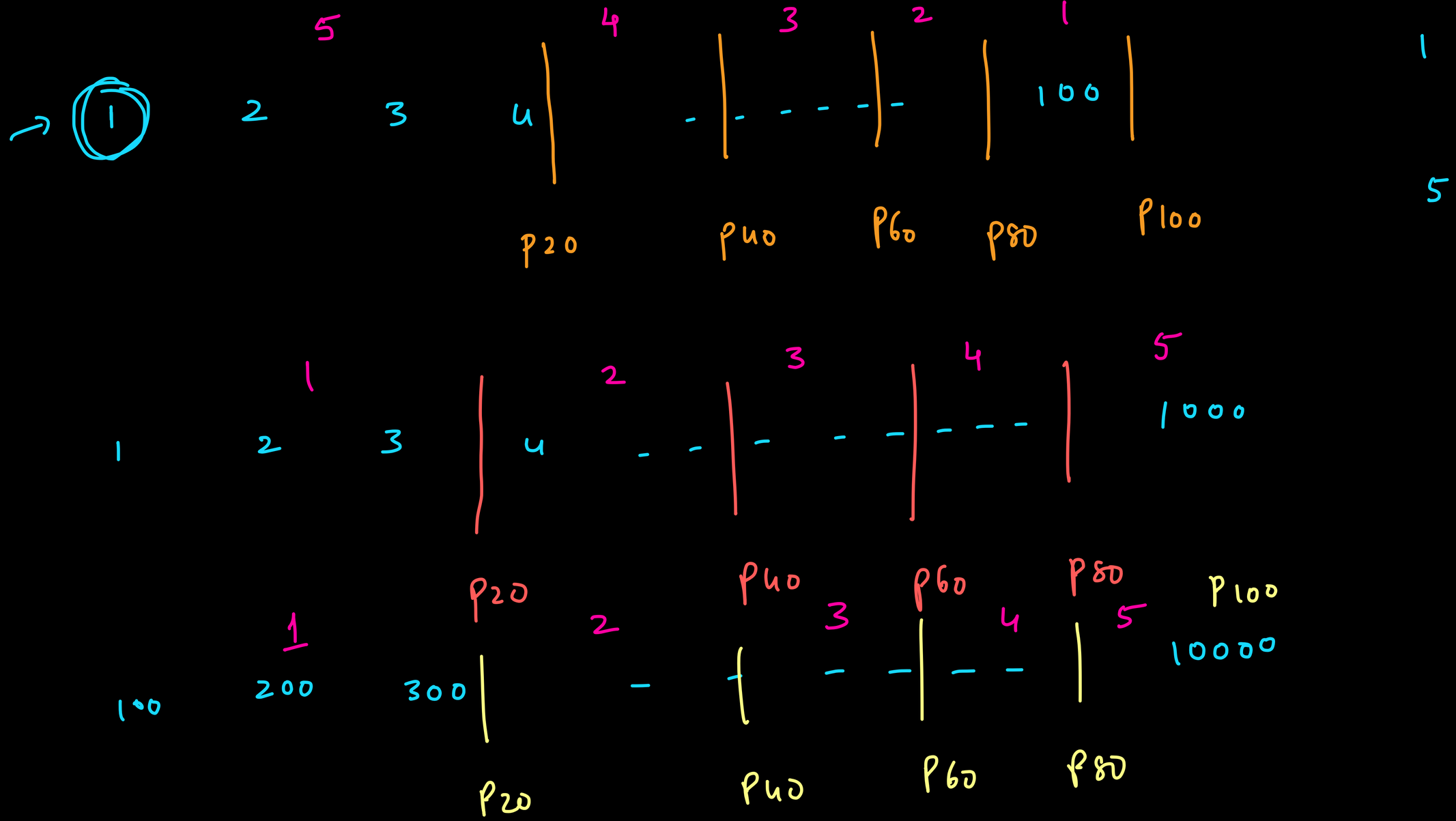
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Recency : How recently customer has made a payment-

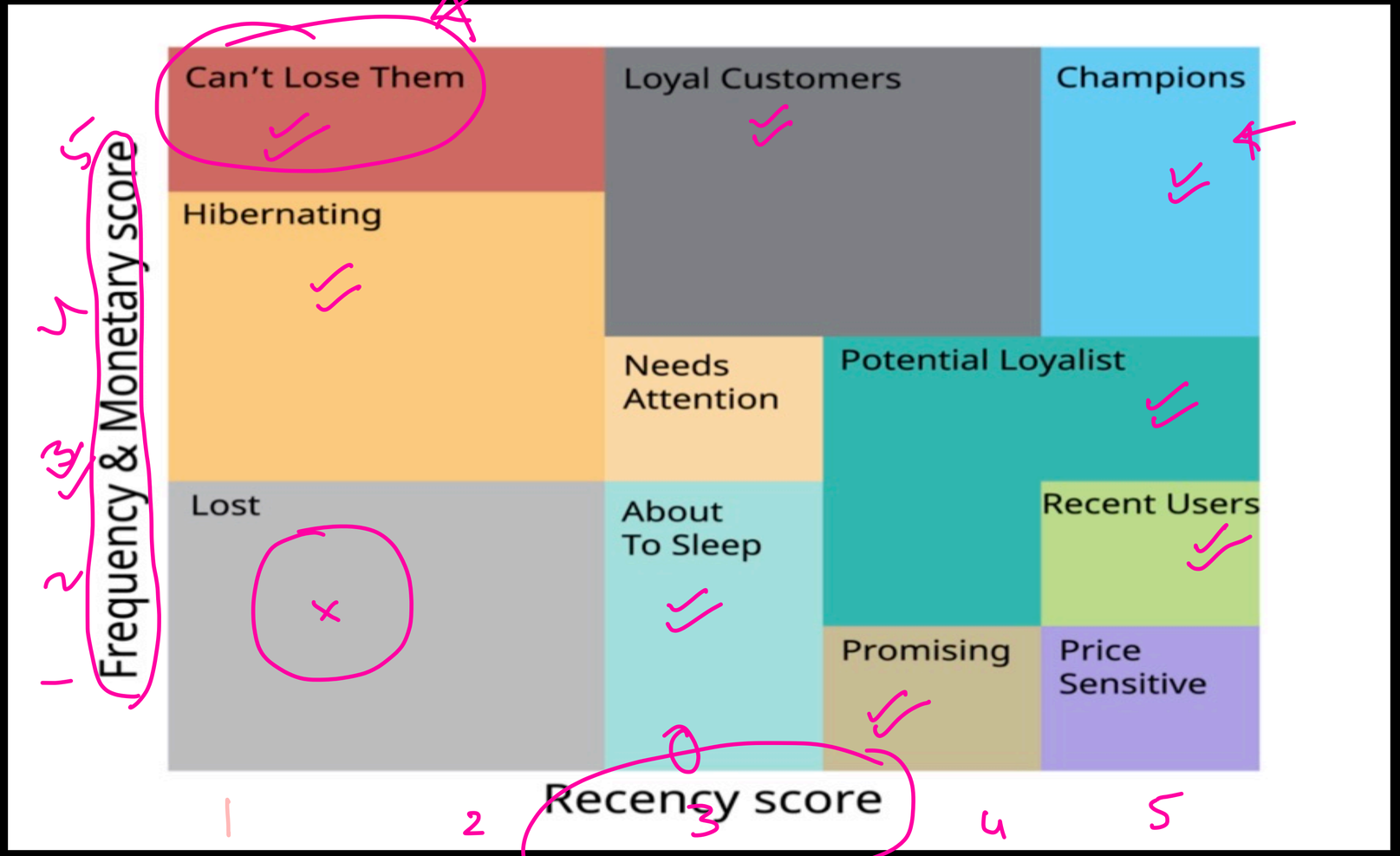
Frequency : How oftem a customer makes a purchase

Monetary : How much amt a customer spends on purchase

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5
4
3
2
1
1
1
2
3
4
100
5
p20
p40
p60
p80
p100
1
2
3
4
5
1000
p20
p40
p60
p80
p100
1
2
3
4
5
100
200
300
10000
p20
p40
p60
p80

Predictive segments



fm_score

5 3

$\frac{5+3}{2} = 4$

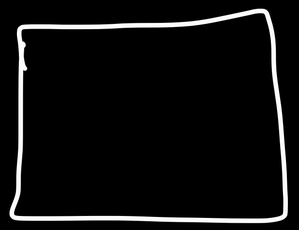
Problem Statement

We will be performing a RFM analysis for a chain of retail stores that sells a lot of different items and categories.

The stores need to adjust their marketing budget and have better targeting of customers so they need to know which customers to focus on and how important they are for the business.

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1. Data processing

customer → RFM → 

2. Compute Recency, Frequency & Monetary value per customer

3. Determine quantiles for each RFM metric

4. Assign scores for each RFM metric

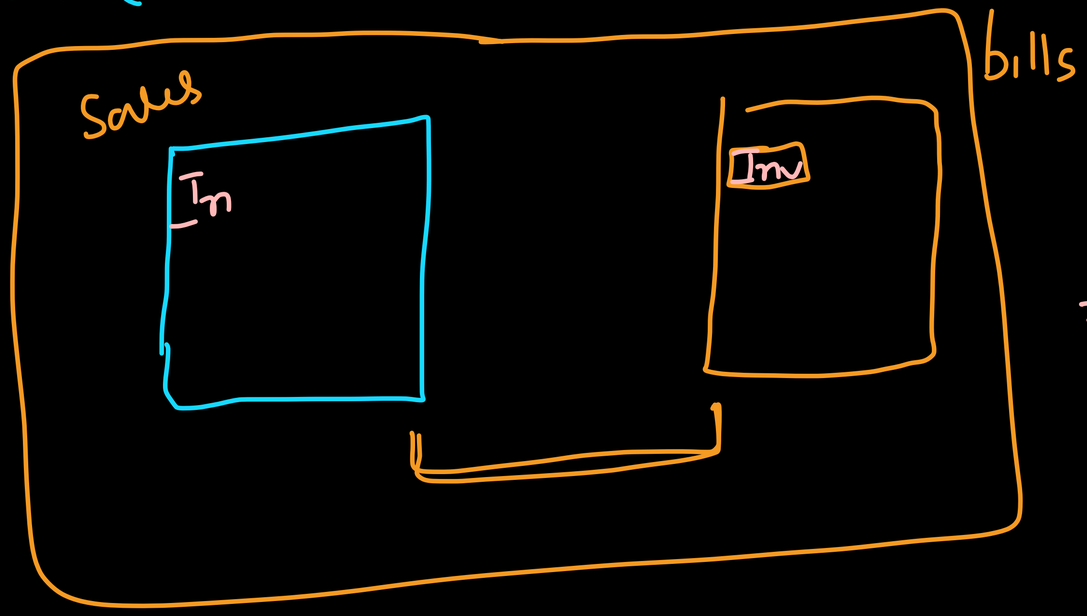
5. Define the RFM Segments using the scores

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1 MAX (Invoice DATE) ⟶ last_purchase date

2 MIN (Invoice DATE) ⟶ first purchase date

3 COUNT (DISTINT Invoice NO) ⟶ Num_purchase



groupby custom

⇒ left join

Recency:

Reference date ← Max (all Invoices)

current

1 + 0

1

2

3

:

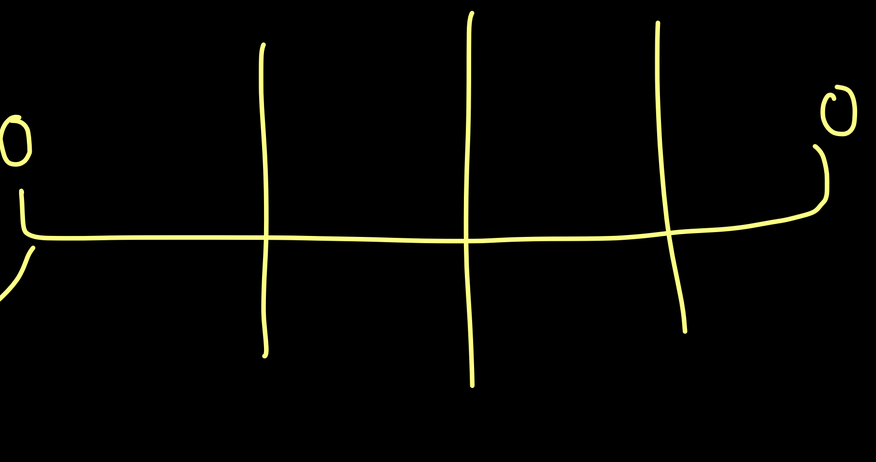
9:59 → 10.10

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Quintile ✓ 1/5 portion of whole

1. APPROX_QUANTILES()

2 OFFSET()



Case when

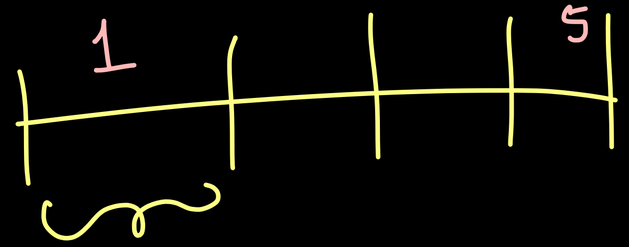
m <=m 20 → 1

>m 20 <=m40 → 2

3

4

5



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